

Yash Shah

6179875489 | syash@umich.edu | [linkedin.com/in/yash-shah1813](https://www.linkedin.com/in/yash-shah1813) | github.com/spicysev-gif

EDUCATION

University of Michigan - Ann Arbor

May 2027

BSE in Computer Engineering + Robotics

3.92/4.0

- **Organizations:** Quant Trading at *Michigan Investment Group*, Microsystems at *Michigan Solar Car Racing*
- **Relevant Coursework:** Data Structures and Algorithms, Circuit Analysis and Design, Digital Signal Processing, Computer Architecture, Discrete Maths, Quantum Mechanics, Linear Algebra, Multivariable Calculus, Differential Equations

Phillips Exeter Academy

Sep. 2021 – Jun. 2025

High School Diploma

PROJECTS

MUREX Robotics Underwater ROV | *Altium, KiCAD, C, Onshape, Fusion 360, Rust, Python*

Sep. 2023 – Present

- Developed end-to-end open source underwater ROV with a 6 DOF frame, 7 DOF arm and 4 fully custom PCBs, that broke 10 industry world-records, proved 88% cheaper than commercial prices and placed 5th in World Competition.
- Created world's Smallest and Cheapest 5 port 100BASE-TX unmanaged ethernet switch from scratch.
- Designed and a thumb-sized, high-data-density smart power module supporting 12 A, 5 V DC-DC with triple-channel power monitoring working at 95% efficiency.
- Developed world's first open source implementation of USB 3.0 in CM4 based carrier board.
- Designed USB-Modular ESC with sensorless FOC using integrated FETs working at 98% efficiency.

Solar Car - World Solar Car Challenge | *Altium, C++*

Oct. 2021 – May 2025

- Designing high-power custom Motor Controller with sensorless FOC for BLDC with 95% efficiency.

Land ROV: Runners Up in First Tech Challenge | *Onshape*

Oct. 2021 – May 2025

- Produced the 1st ever invertible octopod chassis that could strafe in all directions and traverse over rough terrain.
- Designed 5-DOF intake and outtake claws capable of transferring freight and other game elements.

EXPERIENCE

Optiver

March 2026

Future Focus Quants

Chicago, IL

- Gaining insights into the data-driven environment that Optiver traders and researchers work in.

Robotics Engineer

June 2025 – Present

Bedestrian Delivery Robotics

Ann Arbor, MI

- Designed an automated garage and docking mechanism for the B1 Autonomous Robot.
- Developed the LIDAR system and a custom modular tensioned E-Stop Button for the turret system.

Manufacturing Engineer | *Fusion 360 CAM, MasterCAM*

June 2023 – Aug. 2024

Schnell Machine Tools

Thane, India

- Researched DC servo design and achieved 15lb weight reduction in a submarine fin system.
- Conducted FEA structural analysis on automotive suspension systems achieving 30% reduction in weight and 17% increase in load-bearing capacity.
- Prototyped and adapted initial designs using waterjet cutting, 5-axis CNC milling.

HONORS

USAMO Math Competition 1st Quartile

Nov 2023 – March 2024

- Placed in top 230 Students in the World amongst 300000 participants and achieved Distinguished Honor Roll (top 1%) in AMC 12.

#1 on Y-Combinator's Hacker News Forum

June 2024

- Upvoted 666 times on HN and featured on Tom's Hardware, Interesting Engineering and All3DP forums for my development of the World's cheapest and smallest ethernet switch.

TECHNICAL SKILLS

Computer Aided Design: Autodesk Fusion 360, Onshape, Solidworks, Siemens NX, Creo

Printed Circuit Board Design: Altium, KiCAD, SPICE Simulation

Communication Protocols: USB, SPI, I2C, CAN, Ethernet

MicroControllers: Arduino (ATmega328P/2560, SAMD21), ESP32/ESP8266, STM32, Raspberry Pi Pico (RP2040)

Programming: Embedded C, C++, Python, Rust, ROS, MATLAB, Java, MySQL, HTML

Computer Aided Simulation: Ansys Fluent, Autodesk CFD, Ansys Mechanical, Siemens Star-CCM+

Metal Fabrication: Autodesk Fusion 360 CAM, Multi-axis CNC Machining, 3D Printing, Laser Cutting, Waterjet cutting